

```

root@hp-HP-431-Notebook-PC: ~
root@hp-HP-431-Notebook-PC:~# mn
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>

```

Figure 1: Mininet in action

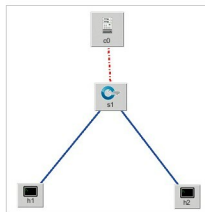


Figure 2: Topology created using Miniedit (an open source tool to create Mininet topologies)

```

mininet> nodes
available nodes are:
h1 h2 s1
mininet> net
h1 h1-eth0:ssi-eth1
h2 h2-eth0:ssi-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
mininet>

```

Figure 3: Nodes and net

```

mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.516 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.104 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.096 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.088 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.098 ms

```

Figure 4: The command makes h1 ping h2

The above command will list all the nodes present in the created network.

As shown in Figure 3, the nodes s1, h1, h2 are displayed.

The command to display and list the links present in the network is:

```
Mininet>net
```

As shown in Figure 3, the interface `eth0` of the host `h1` is connected to `eth1` of switch `s1` and the interface `eth0` of host `h2` is connected to `eth2` of switch `s2`.

The command to display the IP addresses and the process IDs of the nodes is:

```
Mininet>dump
```

As seen in Figure 3, `h1` is assigned the IP address `10.0.0.1` with the process ID `4403` and `h2` is assigned the IP address `10.0.0.2` with the process ID `4407`.

- The command to ping a specific host to a targeted host is:

```
Mininet> h1 ping h2
```

On inspecting the packets, we can see

No.	Time	Source	Destination	Protocol	Length	Info
1.0	0.00000000	0e:33:f6:ed:fc:43	1a:1b:c3:b0:6c:e4	ARP	42	Who has 10.0.0.1? Tell 10.0.0.2
2.0	0.000023051	1a:1b:c3:b0:6c:e4	0e:33:f6:ed:fc:43	ARP	42	10.0.0.1 is at 1a:1b:c3:b0:6c:e4

Figure 5: Packet captured at the interface `s1-eth1`.

```

mininet> h1 ifconfig -a
h1-eth0  Link encap:Ethernet  HWaddr ca:84:bb:23:16:25
         inet addr:10.0.0.1  Bcast:10.255.255.255  Mask:255.0.0.0
         inet6 addr: fe80::c884:bbff:fe23:1625/64 Scope:Link
         UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
         RX packets:30 errors:0 dropped:0 overruns:0 frame:0
         TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
         collisions:0 txqueuelen:1000
         RX bytes:4153 (4.1 KB)  TX bytes:648 (648.0 B)

lo       Link encap:Local Loopback
         inet addr:127.0.0.1  Mask:255.0.0.0
         inet6 addr: ::1/128 Scope:Host
         UP LOOPBACK RUNNING  MTU:65536  Metric:1
         RX packets:0 errors:0 dropped:0 overruns:0 frame:0
         TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
         collisions:0 txqueuelen:1
         RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

```

Figure 6: Getting details of host1 interface

that the first ping has taken considerably longer (0.805) than the others. This is because ARP tables, MAC tables, etc. are initialised during the first ping.

The command to display the address information of the nodes is:

```
Mininet> h1 ifconfig -a
```

This command will display the IP address, broadcast address and MAC address of the host `h1`, as in Figure 6.

- The command to test the connectivity among hosts is:

```
Mininet>pingall
```